publication of the proceedings. In calling attention to the proposed increase of $1 a year in the annual dues, he reminded the members that the value of the proceedings is much more than the full amount of the dues. In referring to future work for the association, Mr. MacFarland suggested the development of a standard of fuel performance for locomotives. At the present time all fuel performance records are comparative with no basic or potential standards and he suggested the investigation of the possibility for the development of a formula for fuel performance in which all the variables are included as factors, as an important subject for the consideration of a standing committee of the association.

Report of Committee on Feed Water Heating

The feed water heaters of today may be divided into the following general classes:

1. Waste gas heaters,
   a. Gas tube,
   b. Water tube.
2. Exhaust steam feed water heaters,
   a. Closed and open types.

The gas tube feed water heaters are used by the various railroads where the Mallet type locomotives have been developed. This type gives a rise in temperature of 70 deg. when new, but in bad water districts this efficiency is lowered rapidly, due to scale forming on the heater surface.

Very little has been accomplished with water tube heaters of the waste gas type, although this type has been the subject of much experimental work. Waste gases passing through the front end will average about 600 deg. in temperature, but the rate at which heat can be transmitted from these gases to the feed water is so low that it necessitates a very large heating surface in the smokebox to realize any appreciable quantity of this heat.

Tests have recently been run on the Santa Fe system with a feed water heater of the Thompson type. After this heater had been four months in service, tests were made. The rise in temperature secured by means of this feed water heater at the end of four months' service was 30 degrees. An examination of the heating surface showed that in the four months' time 1/16 in. of scale had formed on the feed water heating surface. Tests were also made at this time, using a pump in place of the injector, and approximately 70 deg. rise in temperature was secured. No facilities had been designed whereby the scale could be removed from this type of feed water heater, so it was removed after this test was made.

With proper means for the removal of scale and sediment the waste gas heater can be made an effective instrument for the removal of scale forming ingredients and other impurities before these are carried into the boiler.

The exhaust type feed water heaters, due to the lower temperatures, made it practically necessary to use a steam pump in place of the injector. This type is very attractive, due to the fact that the transfer of heat from steam to a heating surface is about ten times that of the transfer of heat by means of waste gases. This allows a rather compact system and the feed water can be brought to within 10 or 15 deg. of the temperature of the exhaust steam.

The closed type of exhaust steam feed water heaters in use in this country are as follows:

Weir feed water heaters, two of which are installed for test purposes on the Southern Railway. The troubles to date with this system have been that the water hammering on the pipes between the pump and feed water heater causes leaks and the expansion and contraction in the feed water heater itself causes leaks between the copper tubes and cast iron tube plates. A re-designed feed water heater has been planned so as to take care of this latter trouble.

The second feed water heater system of this type developed in this country is the heater manufactured by the Locomotive Feed Water Heater Company. This heater has been in service on the New York Central, the Delaware & Hudson, and the Delaware, Lackawanna & Western railways, and to date in the current year the following applications have been or are being made: Central Railroad of New Jersey, Fort Smith & Western, Canadian Pacific, Grand Trunk and Erie railroads. Additional equipments are being applied to the Delaware & Hudson and the New York Central. The design is made so that the expansion and contraction of the brass tubes is taken care of so that no leakage occurs.

In order to increase the effectiveness of the heating surface in the brass tubes a thin brass, corrugated spiral agitator is placed in each tube. This agitator more than doubles the transfer of heat across each square foot of heating surface, and the high velocity and scouring action of the water caused by its presence tends to prevent lodgment of scale or dirt on the heating surface. Not the slightest particle of scale or dirt has ever been found in the heaters now in service. No experience, however, has been obtained from very bad waters.

The open type heater in service is the McBride feed water heater and pump, manufactured by the Worthington Pump & Machinery Corporation. There are four in service on the Pennsylvania Lines East on locomotives in service out of Altoona, Pennsylvania. These have been in service for about a year. The feed water heater and pump are combined and are designed for a capacity of 60,000 lb. of feed water per hour.

The exhaust steam passes through an oil separator enroute to the feed water heater. No trouble has been experienced so far from oil passing through the oil separator with the exhaust steam and being delivered to the boilers with the feed water. On road tests the increase in evaporation per pound of coal was 14.4 per cent for the heater.

The development of the locomotive feed water heater is not a question of theory, but of design. A locomotive feed water heater must be capable of continuous operation under the most adverse conditions; it must not add appreciably to the difficulty of controlling water supplied to the boiler at any required rate; it should not increase the weight on driving wheels to the extent that it becomes necessary to reduce the dimensions of the boiler, and it should be applicable to existing locomotives without necessitating radical alterations.

The substitution of a pump for an injector is the most difficult problem connected with the feed water heating. It complicates the control of boiler feed, increases locomotive maintenance and weight, and is in itself not as efficient nor reliable an instrument as the injector. On this account the success of locomotive feed water heating hinges very largely on the design of a successful pump, unless a practical heater of the waste gas type can be designed which will raise feed water temperatures sufficiently to effect an appreciable saving with the use of an injector.

No great progress in feed water heating can be expected until the solution of this problem is actively taken in hand by the railroads with a determination to make feed water heating a practical operating factor. This committee would urge that the mechanical department of each railroad endeavor to lead in this development. We should each endeavor to be a pioneer in the practical application of the feed water heater.

The report was signed by E. E. Chapman (chairman), A., T. & S. Fe.; E. A. Averill, Loco, Feed Water Co.; O. S. Beyer, Jr.; E. V. De Vilbiss, Penn. System; J. S. Hampson, Rosengrant Coal Co.; F. Kerby, B. & O.; Monro B. Lanier, Monroe-Warrior Coal & Coke Co.; L. G. Plant, *Railway Age*, and L. R. Pyle, Loco, Firebox Co.

Discussion

E. A. Averill (Locomotive Feed Water Heater Company) stated that there are three essential considerations which must be taken into account in judging the practicability of feed